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**Softwarový vývoj**


**EXTENDED SUMMARY**

# Modeling the spread of opinions in social networks


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# Introduction, aim, and research questions

Opinion dynamics studies how interactions between individuals can produce collective states such as consensus, polarization, or fragmentation. Agent-based models isolate mechanisms of social influence and examine their consequences under controlled conditions. Their results depend both on the opinion-update rule and on the social network through which interactions take place. Different topologies may therefore provide different opportunities for opinions to converge or remain separated.

This thesis focuses on continuous bounded-confidence models, in which actors are influenced only by opinions within a confidence threshold. Two established models are examined: the Deffuant model, which updates a randomly selected pair of connected actors, and the Hegselmann–Krause (HK) model, in which actors synchronously move toward the mean opinion of acceptable neighbors.

The principal aim is to implement these models in Julia and determine how their parameters and network topology affect the resulting opinion distribution. They are compared on an Erdős–Rényi random network, a Barabási–Albert scale-free network, and an observed network of politically oriented Facebook pages. The latter supplies a real topology, not a direct representation of interpersonal influence or the United States Congress.

Three research questions guide the investigation: How does the confidence threshold affect the transition between fragmentation and consensus, and does this transition differ between the two update rules? Do networks of comparable size and average degree produce different outcomes when their edge arrangements differ? Can local structural information be incorporated into the Deffuant model in a way that qualitatively changes its dynamics?

The third question led to the Heterogeneous Resistance Deffuant model with Community Reinforcement (HRDCR). Its construction hypotheses assume that neighborhood overlap may modify the effective confidence threshold, common community membership may reinforce an accepted interaction, and highly connected nodes may resist opinion change. These are not presented as empirical laws; the model analyzes their computational consequences.

# Methods

All three models were implemented in Julia. Each network contained 5,910 nodes, and initial opinions were independently sampled from a uniform distribution on the interval from zero to one. Every parameter configuration was repeated 100 times. The confidence threshold took values from 0.1 to 0.4; the Deffuant and HRDCR convergence parameter used the same range. A Deffuant or HRDCR run contained one million random edge selections. HK used 338 synchronous iterations, making the potential number of node updates approximately comparable across the baseline models.

Three measures were evaluated jointly. DBSCAN identified dense opinion clusters, using a radius of 0.01 and a minimum cluster size of one percent of the population. Pairwise average distance (PAD) measured overall dispersion. Polarization was estimated with the Duclos–Esteban–Ray index after Gaussian kernel density estimation. Stored intermediate states supported trajectory analysis. Selected results were also compared with first-dimension NOMINATE scores from Voteview. This supplied empirical context, not individual calibration or predictive validation.

## Key sources

The following eight sources were the most important for the theoretical framework, model definitions, evaluation, and empirical context of the thesis:

1. **Castellano, Fortunato, and Loreto (2009), "Statistical Physics of Social Dynamics."** Provided the general framework connecting local interaction rules with collective social outcomes.
2. **Flache et al. (2017), "Models of Social Influence: Towards the Next Frontiers."** Supported the interpretation and limitations of agent-based social-influence models.
3. **Deffuant et al. (2000), "Mixing Beliefs among Interacting Agents."** Defined the pairwise baseline and the foundation of HRDCR.
4. **Hegselmann and Krause (2002), "Opinion Dynamics and Bounded Confidence."** Defined the synchronous baseline and enabled comparison with pairwise updating.
5. **Lorenz (2007), "Continuous Opinion Dynamics under Bounded Confidence."** Clarified model variants and the role of confidence in cluster formation.
6. **Barabási (2016), *Network Science*.** Supplied the background for network structure, clustering, paths, and scale-free topology.
7. **Duclos, Esteban, and Ray (2004), "Polarization: Concepts, Measurement, Estimation."** Introduced the continuous polarization measure used in the experiments.
8. **Lewis et al. (2026), Voteview.** Supplied nominate_dim1 observations for empirical context; Poole's spatial-voting methodology guided their interpretation.

## Main results and original contribution

The confidence threshold was the strongest control parameter in both baseline models. Low values preserved separated groups; higher values admitted more interactions and moved both models toward consensus. The pairwise Deffuant transition was gradual, while synchronous HK changed more sharply. On the real network, the mean HK cluster count fell from 8.37 at a threshold of 0.1 to 1.13 at 0.2; Deffuant declined from 6.62 to 2.66. Thus, bounded confidence alone does not determine the outcome: the update rule also matters.

Network topology also mattered. At an HK threshold of 0.1, the real network produced 8.37 clusters, compared with approximately 2.75 on both synthetic networks. Deffuant showed a smaller but similar difference. Because node counts and average degrees were comparable, average connectivity alone cannot explain the result. Detailed edge arrangement influenced repeated exchange and the persistence of local groups.

No single measure was sufficient. One DBSCAN cluster could mean narrow consensus or a continuous opinion band with little convergence. PAD distinguished concentrated from dispersed distributions. Conversely, near consensus, kernel density estimation could create a narrow, high peak and a large numerical polarization value despite low PAD. Cluster count, PAD, and polarization therefore had to be interpreted together.

The principal original contribution is the design, implementation, and evaluation of HRDCR. Heterogeneous resistance makes node inertia degree-dependent. Community reinforcement modifies effective confidence through Jaccard neighborhood overlap and strengthens updates within communities. These mechanisms operationalize the stated construction hypotheses.

HRDCR qualitatively changed the baseline dynamics. On the real network at a confidence threshold of 0.4, it produced 7.67 clusters and a PAD of 0.309. Deffuant produced 1.055 clusters and a PAD of 0.046, while HK produced one cluster with a PAD of 0.018. HRDCR thus favored local grouping while preventing local interactions from becoming population-wide agreement.

This is not evidence that HRDCR is universally better or more realistic. Its mechanisms were not calibrated on longitudinal observations, four structural parameters remained fixed, and their effects cannot be separated without ablation. The contribution is a transparent demonstration that local network information can alter bounded-confidence dynamics.

The NOMINATE comparison provided a limited external reference. The 119th Senate distribution had a PAD of 0.272 and polarization of 0.413. Across the three measures, HRDCR on the real network at a threshold of 0.4 was the closest tested configuration. It does not reproduce Senate behavior: nodes were not matched to legislators and the simulation did not begin from NOMINATE values. Its broad output merely resembled the aggregate reference more closely than baseline consensus at high thresholds.

The assignment was fulfilled through a reproducible Julia implementation, systematic comparison of two established models across three topologies, and joint evaluation by several measures. HRDCR and the supplementary congressional comparison extended rather than replaced the baseline analysis. The observed dependence on topology motivated introducing local structure directly into the interaction rule.

Further work should conduct sensitivity and ablation analyses, record successful interactions, and use a convergence condition alongside a fixed horizon. Robustness should be tested with alternative cluster tolerances, bandwidths, initial distributions, and observed networks. Empirical validation ultimately requires a network linked to repeated opinion measurements of the same actors, allowing resistance and structural effects to be calibrated rather than assumed.